

ILLUSTRATED ASSEMBLY GUIDE

QIDI Plus4 optimized print package
305 x 305 x 280 mm nominal envelope

BUILD ARCHITECTURE

610 x 457 x 18 mm precut board

3 indexed printed stations

4 locator pins + 9 M4 retainers

Direct-applied T0-T3 / K0 / P0 tags

No router. No CNC.

RELEASE STATUS: UNRELEASED

Digital geometry and document checks do not constitute physical release.

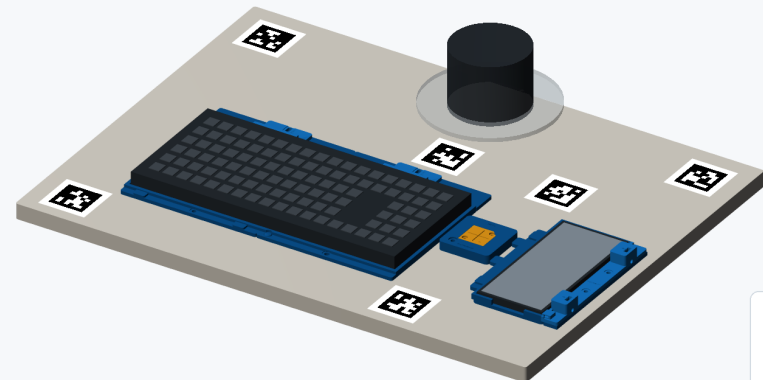
Build, measure, proof-load, commission, record, and sign every acceptance gate.

15

FINAL ASSEMBLY CHECK

RC03-INT-R1

Verify every station is seated, every device is gentle and every datum is visible.



1

Black rear cylinder = RoArm keep-out proxy, not released clamp CAD

2

Run 10x station remove/reinstall cycles; calibrate board + TCP

3

Run empty-cell motion before keyboard or phone contact

Stop before fabrication if any box is open

UNRELEASED UNTIL PHYSICAL ACCEPTANCE IS RECORDED

The drawings and renders communicate intent. They cannot prove fit, strength, motion clearance, or camera performance.

BEFORE DRILLING OR INSERTS

- ☐ BOM, FASTENER_MAP.csv, and revision agree
- ☐ Board is 610 x 457 x 18 mm and secured against shift
- ☐ Board is flat, sealed equally on both faces, and fully cured
- ☐ All fit coupons passed after cooling
- ☐ M4 anchor type and final bore were selected on the 18 mm cutoff test
- ☐ Arm reinforcement plate is measured metal and fits the factory clamp load path
- ☐ E-stop is mounted outside robot reach
- ☐ Power is locked out for drilling, insert installation, and measurement
- ☐ 1:1 template scale has been verified in X and Y
- ☐ No router or CNC operation is planned or required

NO-GO CONDITIONS

- ☐ Do not drill direct AprilTag sites
- ☐ Do not use printed parts as the arm structural load path
- ☐ Do not force locator pins or ream production parts to hide a fit error
- ☐ Do not tighten screws before every shared stack is fully seated
- ☐ Do not power motion with a device or tool installed during the empty-cell test
- ☐ Do not accept bottomed screws, underside projection, rocking, or clamp distortion
- ☐ Do not treat the rendered arm cylinder or camera view as measured clearance
- ☐ Do not release from CAD checks alone

Core parts and load-path hardware

Stage items into labeled bags. A checkmark means the exact quantity and specification were verified.

CORE

- ☐ 1 x Waveshare RoArm-M3 Pro
- ☐ 1 x Structural board
- ☐ 1 x Compact keyboard
- ☐ 1 x Samsung Galaxy A16 5G

BOARD INTERFACE

- ☐ 6 x 6 mm precision dowel pins (4 installed + 2 spares)
- ☐ 12 x M4 board threaded interfaces (9 installed + 3 spares)
- ☐ 10 x M4 station retention screws (7 installed + 3 spares)
- ☐ 12 x M4 station washers (9 installed + 3 spares)
- ☐ 1 x Slow-cure two-part epoxy

ARM INTERFACE

- ☐ 1 x Precut metal underside reinforcement plate

ASSEMBLY

- ☐ 2 x Keyboard rear-clamp face pad pieces
- ☐ 4 x M4 low-profile keyboard clamp/retention screws (2 installed + 2 spares)

PHONE CLAMP

- ☐ 4 x M4 captured hex nuts (2 installed + 2 spares)
- ☐ 2 x M4 x 25 mm thumb screws or nylon screws

TCP RECEIVER

- ☐ 4 x M3 heat-set inserts (2 installed + 2 spares)
- ☐ 4 x M3 x 10 mm button-head screws (2 installed + 2 spares)

COMPLIANT TOOL

- ☐ 2 x M3 heat-set inserts
- ☐ 2 x M3 x 10 mm button-head screws
- ☐ 1 x Compression spring

KEYBOARD TOOL

- ☐ 1 x 6 mm smooth rod - One tool option

PHONE TOOL

- ☐ 1 x Passive capacitive stylus - One tool option

TOOL RETENTION

- ☐ 1 x Removable retaining compound - If needed

SAFETY

- ☐ 1 x Latching power cutoff / E-stop

Fabrication, vision, cable, and optional support

Recommended and optional lines remain labeled as such. Do not substitute required measurement or safety tools.

FABRICATION

- [] 1 x Digital calipers
- [] 1 x Heat-set insert installation tip
- [] 1 x Hand drill and depth-control set
- [] 1 x Filament dryer or dry storage - Recommended
- [] 1 controlled lot x QIDI ABS Rapido filament
- [] 1 qualified spool x PETG filament for split adapters - Route dependent
- [] 1 qualified spool x TPU 95A filament for contact tips
- [] 1 qualified spool x ASA filament for fixed-mast fallback - Optional
- [] 1 x Digital force gauge or calibrated weights
- [] 1 x Straightedge and feeler gauge set
- [] 1 x Low-range torque driver

VISION

- [] 1 x USB camera
- [] 2 x Matte white paper/card
- [] 12 x Full-surface matte adhesive AprilTag tiles (6 installed + 6 spares)
- [] 1 x Lint-free wipes and finish-compatible cleaner

CABLE MANAGEMENT

- [] 12 x Nylon cable ties
- [] 4 x Adhesive cable-tie anchors - Recommended

CAMERA STAND

- [] 2 x 2020 extrusion - Optional
- [] 1 x 2020 extrusion - Optional
- [] 4 x 2020 angle brackets + T-nuts - Optional
- [] 4 x M5 x 60 mm clamp bolts + nuts - Optional
- [] 8 x M5 base-mounting hardware - Optional
- [] 2 x M5 T-slot screws - Optional
- [] 2 x M5 T-nuts - Optional
- [] 16 x M5 flat washers - Optional
- [] 4 x Camera retention straps or project-standard zip ties - Optional
- [] 1 x 1/4-20 camera screw and nut - Optional

BOARD

- [] 6 x Rubber feet - Recommended
- [] 1 x Matte neutral paint or vinyl - Recommended

SERVICE SPARES

- [] 1 x Fastener and insert spare pack - Recommended

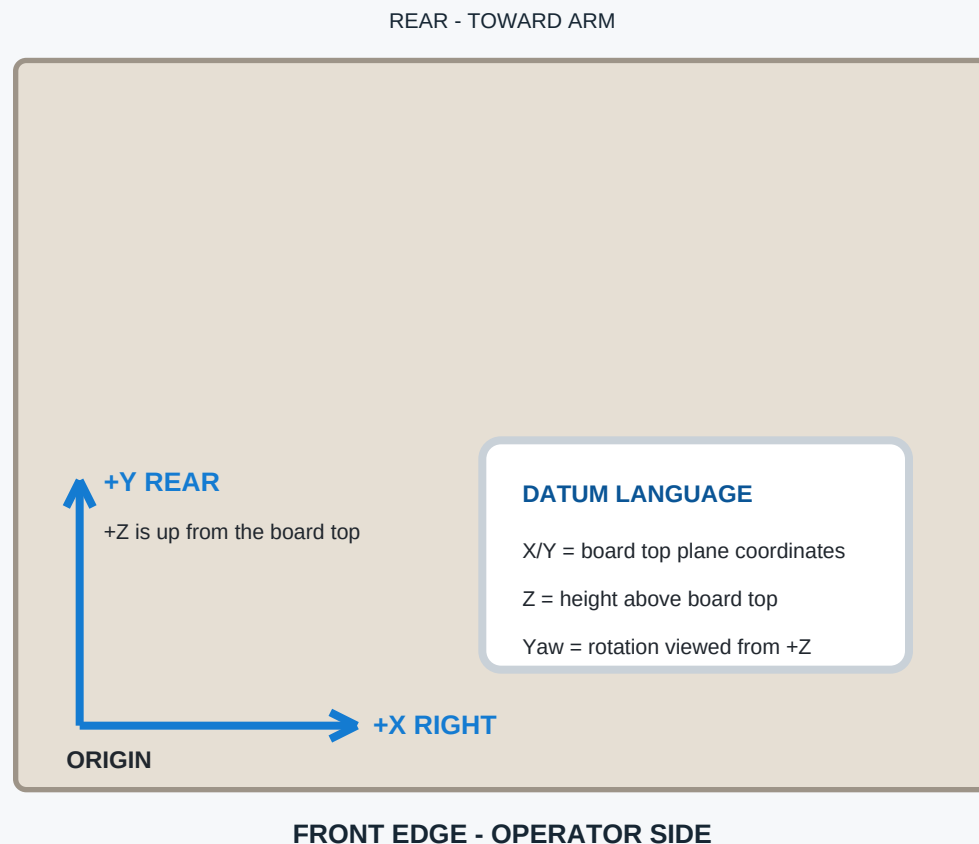
Read this page before placing any station

ILLUSTRATION COLOR KEY

	Blue	Install, place, or act on this item now
	Gray	Already installed or contextual geometry
	Orange	Hardware, adhesive, cable, or motion
	Teal	Compliant or soft-contact component
	Red	Stop, no-go, collision, or release warning

INTERFACE RULES

- ☐ Pins locate. Screws clamp.
- ☐ Keyboard left is the master. Keyboard right is located by the seam.
- ☐ Seat shared rails and sliders before installing their screws.
- ☐ Direct tags are adhesive features - never drilled features.
- ☐ The setup template transfers centers and direction only.



Print and qualify in this order

PRINT ONLY A SELECTED JOB MARKED READY. Keep every accepted coupon with its job, profile, spool lot, and date.

A**QUALIFY PROCESS**

- ☐ Confirm the installed 0.4 mm nozzle.
- ☐ Calibrate each exact dry PETG, TPU, and selected ASA profile.
- ☐ Print required diagnostics 00A, 00B, 00C, 00E, and 00F at 100 percent with their assigned profiles; add 00D only for a selected phone-stylus route.
- ☐ Cool completely before measuring; record evidence with the supplied script.
- ☐ Retain the accepted coupons as controlled setup tools.

B**RELEASE CORE PRINTS**

- ☐ Print and accept job 01 keyboard master first.
- ☐ Use the actual master posts with the retained seam ladder; release job 02 only after the seam gate passes.
- ☐ Then qualify 03A station, 03B sliders, 03C1 rail, 03C2 setup tool, and two 03D cartridges; keep 03C3 blocked pending camera-architecture alignment.
- ☐ Print 05A TPU first article before the 05B phone-tip batch.
- ☐ Do not drill the final board until all three stations are accepted.

C**SELECT ROUTES**

- ☐ Shared compliant body: job 04A after at least one tool route is selected.
- ☐ Phone route: 04B measured stylus collars.
- ☐ Keyboard route: 04C rod bushings, then 05C first article and 05D batch.
- ☐ Fixed-mast fallback: keep jobs 06/07A/07B blocked until the arm-camera decision is `ALIGNED_AND_REVISED` and that same controlled decision explicitly authorizes the fallback; then qualify it physically.
- ☐ Never globally scale or auto-orient a released production plate.

STOP: mesh PASS, successful slicing, or visual fit is not a physical coupon or first-article PASS.

Board fabrication - detailed order

No router. No CNC. Remove the paper before final drilling; use positive depth stops and the accepted hardware route.

1

FINISH AND QUALIFY

- ☐ 610 x 457 x 18 mm board; mark front-left origin, +X right, +Y rear.
- ☐ Seal both faces and all edges equally; cure fully.
- ☐ PASS: local gap ≤ 0.50 mm/300 mm; overall bow ≤ 1.50 mm.

2

REGISTER AND TRANSFER

- ☐ Print the template at Actual Size / 100 percent.
- ☐ Both 100 mm bars must measure 100.0 ± 0.2 mm.
- ☐ Transfer 4 locator and 9 anchor centers; tag outlines are not holes.

3

DRILL CONTROLLED BORES

- ☐ Drill four 6 mm locator bores 15 mm blind with a positive stop.
- ☐ Prepare nine anchor bores with the exact method proven in an 18 mm cutoff.
- ☐ The 4.60 mm drawing symbol is not a universal final anchor bore.

4

INSTALL AND PROVE ANCHORS

- ☐ Require ≥ 5 mm thread engagement with no bottoming or underside projection.
- ☐ Cycle three times at 0.25-0.35 N m.
- ☐ Proof every anchor at 50 N axial load for 60 seconds.

5

DRY-FIT FULL STACK

- ☐ Insert four pins dry at 5.0 mm projection; use no station shims.
- ☐ Seat all three stations plus sliders and phone rail.
- ☐ Start all nine washer/screw stacks; complete three remove/reinstall cycles.

6

BOND ONLY THE PINS

- ☐ Remove every printed station after the dry fit passes.
- ☐ Apply sparse slow-cure epoxy only to the four board pins.
- ☐ Hold 5.0 mm projection; keep adhesive out of printed sockets; cure fully.

7

RECHECK REGISTRATION

- ☐ Each master station must drop on and lift off by hand.
- ☐ STOP for binding, rocking, whitening, forced pins, or screw bottoming.
- ☐ Record the cured pin projection and station seating evidence.

8

PROTECT THE UNDERSIDE

- ☐ Confirm anchors and reinforcement will clear all feet.
- ☐ Install anti-shift feet only after underside clearance is proven.
- ☐ Continue to illustrated Step 02 for the measured metal plate and factory arm clamp.

STOP: do not proceed to production assembly until the complete dry fit, anchor proof, epoxy cure, and post-cure drop-on checks pass.

Authoritative nine-row M4 map

Do not select screw length from the render. Select it on the actual stack and 18 mm cutoff test.

Every row requires >=5 mm engagement, no bottoming, no underside projection, one washer, and 0.25-0.35 N m final torque.

ID	Station	Role	Stack	Hardware class	Candidate	Selected	Acceptance / sequence
KBL-HOLD-F	keyboard left	fixed station retainer	4.0 mm	conventional M4 button head	M4x20 or M4x25	____ mm []	>=5 mm thread engagement; no bottoming or underside projection; 0.25-0.35 N m
KBL-HOLD-R	keyboard left	fixed station retainer	4.0 mm	conventional M4 button head	M4x20 or M4x25	____ mm []	>=5 mm thread engagement; no bottoming or underside projection; 0.25-0.35 N m
KBL-CLAMP	keyboard left	shared rear-clamp and station retainer	8.0 mm	low-profile hand-adjustable M4	M4x20 or M4x25	____ mm []	slider installed before screw; position to padded contact only; 0.25-0.35 N m after seating
KBR-HOLD-F	keyboard right	fixed relieved slave retainer	4.0 mm	conventional M4 button head	M4x20 or M4x25	____ mm []	seam owns XY/yaw; no bottoming or underside projection; 0.25-0.35 N m
KBR-HOLD-R	keyboard right	fixed relieved slave retainer	4.0 mm	conventional M4 button head	M4x20 or M4x25	____ mm []	seam owns XY/yaw; no bottoming or underside projection; 0.25-0.35 N m
KBR-CLAMP	keyboard right	shared rear-clamp and relieved slave retainer	8.0 mm	low-profile hand-adjustable M4	M4x20 or M4x25	____ mm []	slider installed before screw; seam fully seated before tightening; 0.25-0.35 N m
PT-HOLD-TCP	phone tcp	TCP-island recessed-washer station retainer	9.0 mm	conventional M4 button head	M4x25	____ mm []	seat washer fully in 1.5 mm recess; verify >=5 mm engagement; start before rail-side screws; no bottoming or underside projection; 0.25-0.35 N m
PT-HOLD-R1	phone tcp	shared phone-rail and station retainer	8.0 mm	conventional M4 button head	M4x20 or M4x25	____ mm []	rail seated on keys before screw; no bottoming; 0.25-0.35 N m
PT-HOLD-R2	phone tcp	shared phone-rail and station retainer	8.0 mm	conventional M4 button head	M4x20 or M4x25	____ mm []	rail seated on keys before screw; no bottoming; 0.25-0.35 N m

[] 9 / 9 anchors installed [] Shared sliders and rails seated first [] Final torque witnessed

Bag labels must use the feature IDs shown above

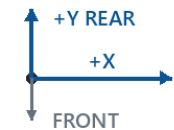
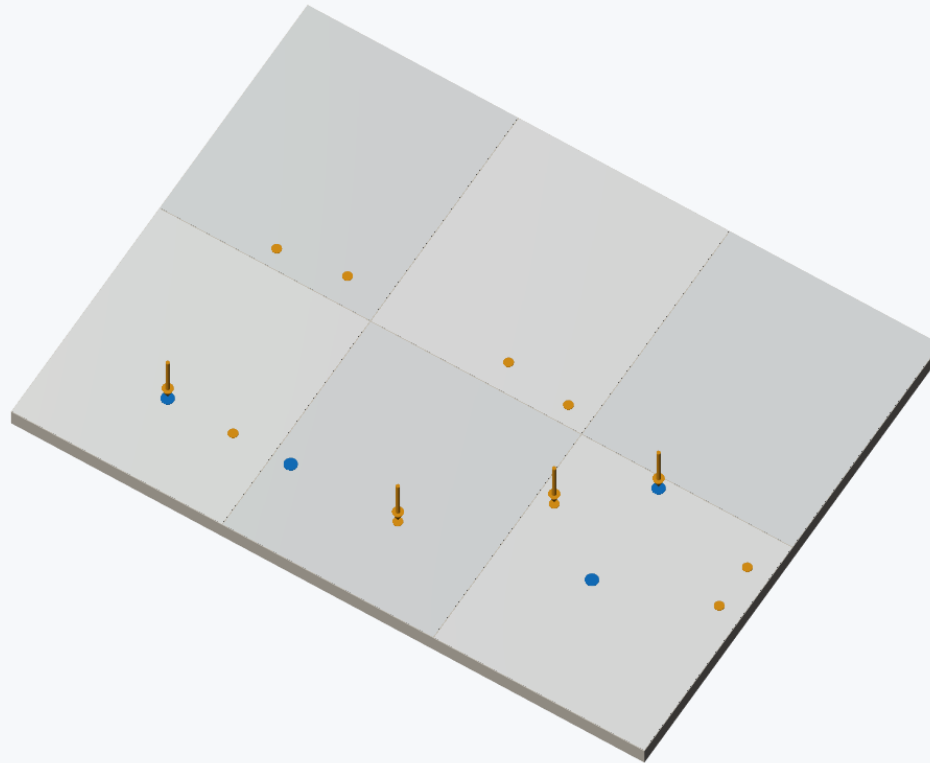
01 / 15 PREPARE THE BOARD

01

PREPARE THE BOARD

NO ROUTER / NO CNC

After cure and flatness acceptance, tape the verified 1:1 template and transfer centers.



1

Seal both faces; cure fully; verify flatness before layout

2

Blue: 4x 6 mm blind locator centers

3

Orange: 9x M4 anchor centers; final bore follows selected anchor + scrap test

☐ STEP ACCEPTED

Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

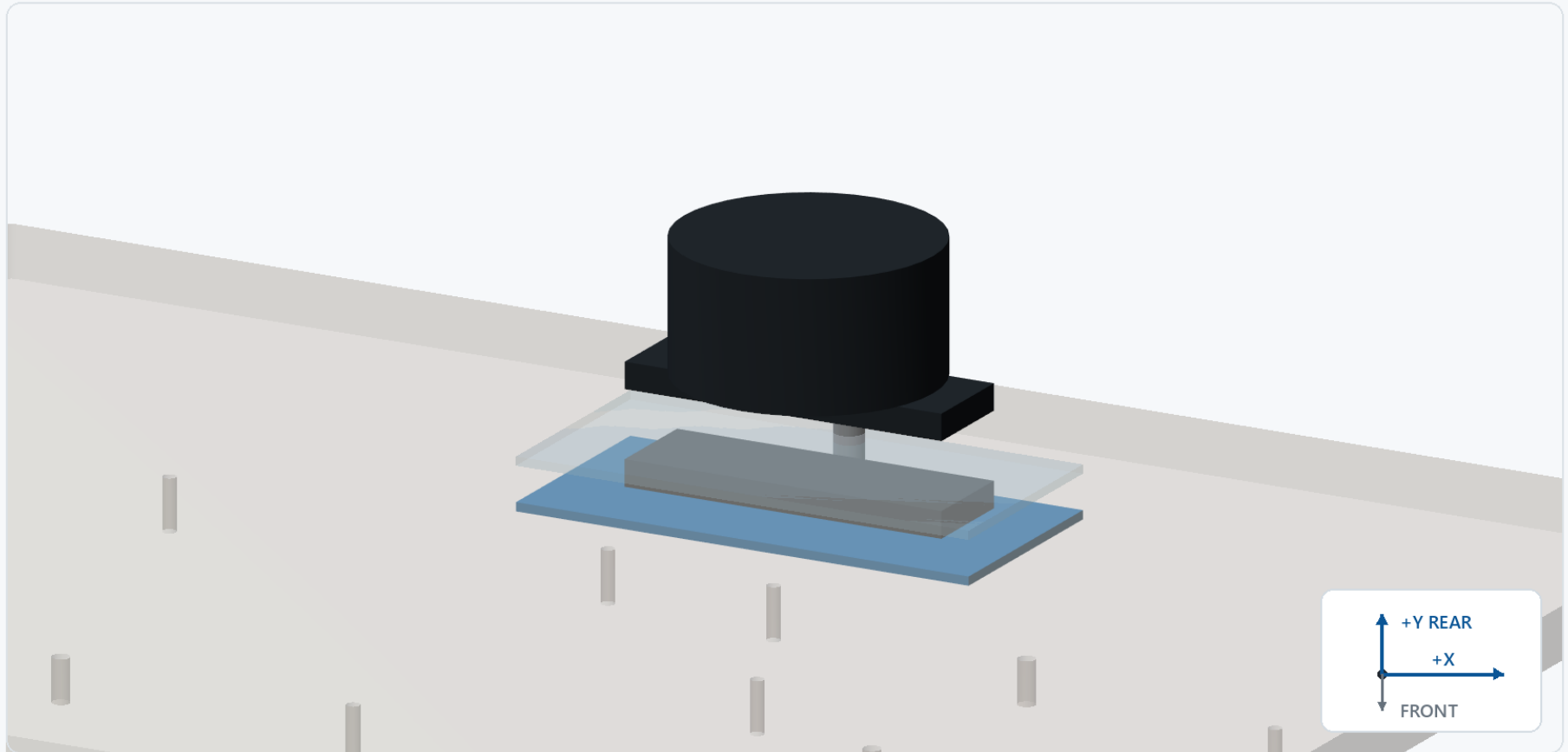
02 / 15 REINFORCE AND CLAMP THE ARM

02

REINFORCE + CLAMP THE ARM

Fit measured precut metal under the board, then use the factory load-bearing clamp.

FACTORY CLAMP IS SCHEMATIC



1

Plate: measured footprint; minimum 3 mm aluminum or 2 mm steel

2

Keep plate clear of feet, anchors and station hardware

3

Factory clamp remains load bearing; verify the board stays flat

[] STEP ACCEPTED

Measurement / photo ID: _____

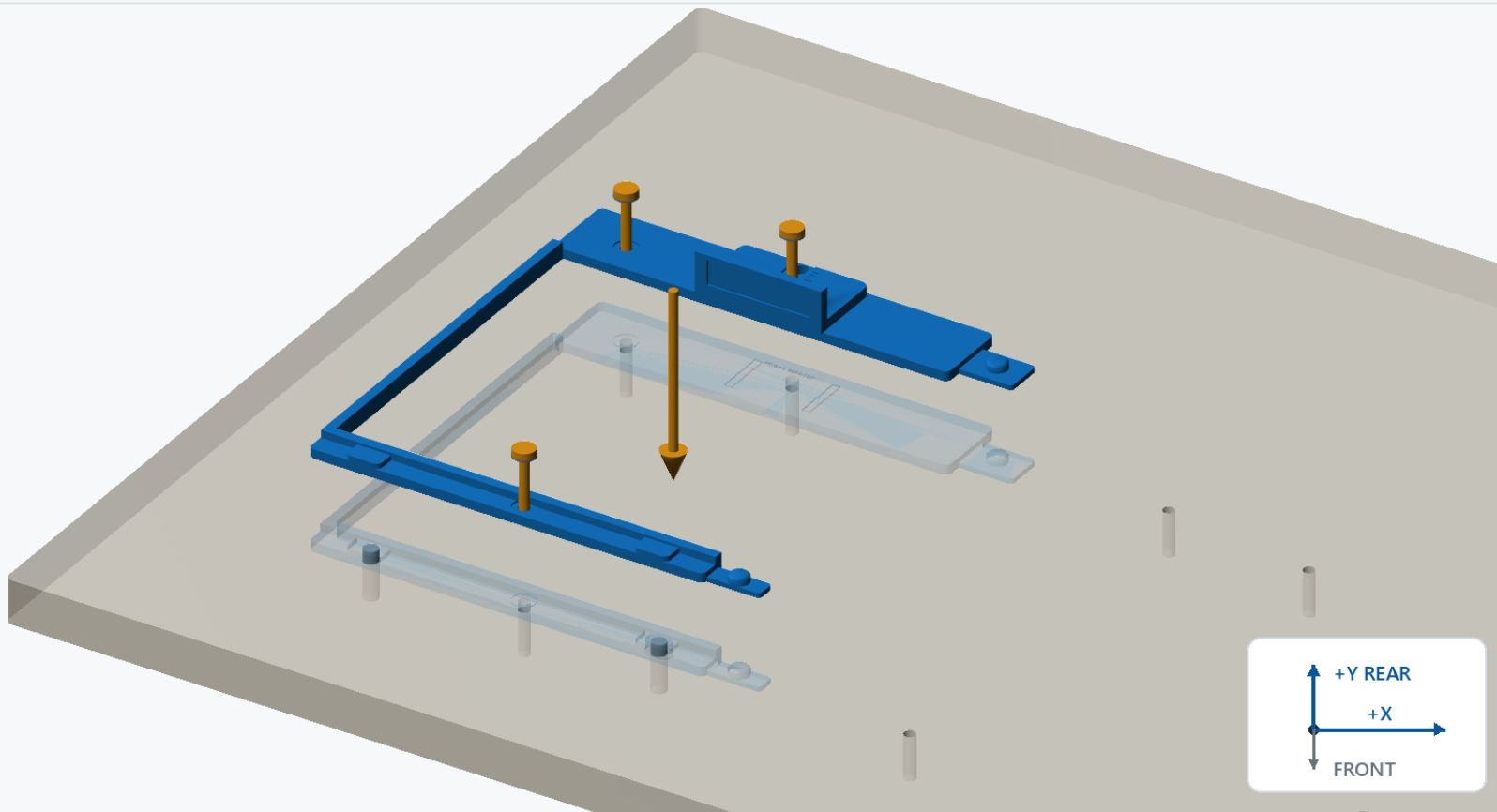
If a gate fails: stop, correct, and repeat this step

03 / 15 INSTALL KEYBOARD MASTER

03

INSTALL KEYBOARD MASTER

Seat the left station on its round and radial locator pins.



1

2x locator pins set master position; lower squarely

2

Start front + rear M4/washer retainers loosely

3

Seat padded slider before the shared third M4/washer retainer

[] STEP ACCEPTED

Measurement / photo ID: _____

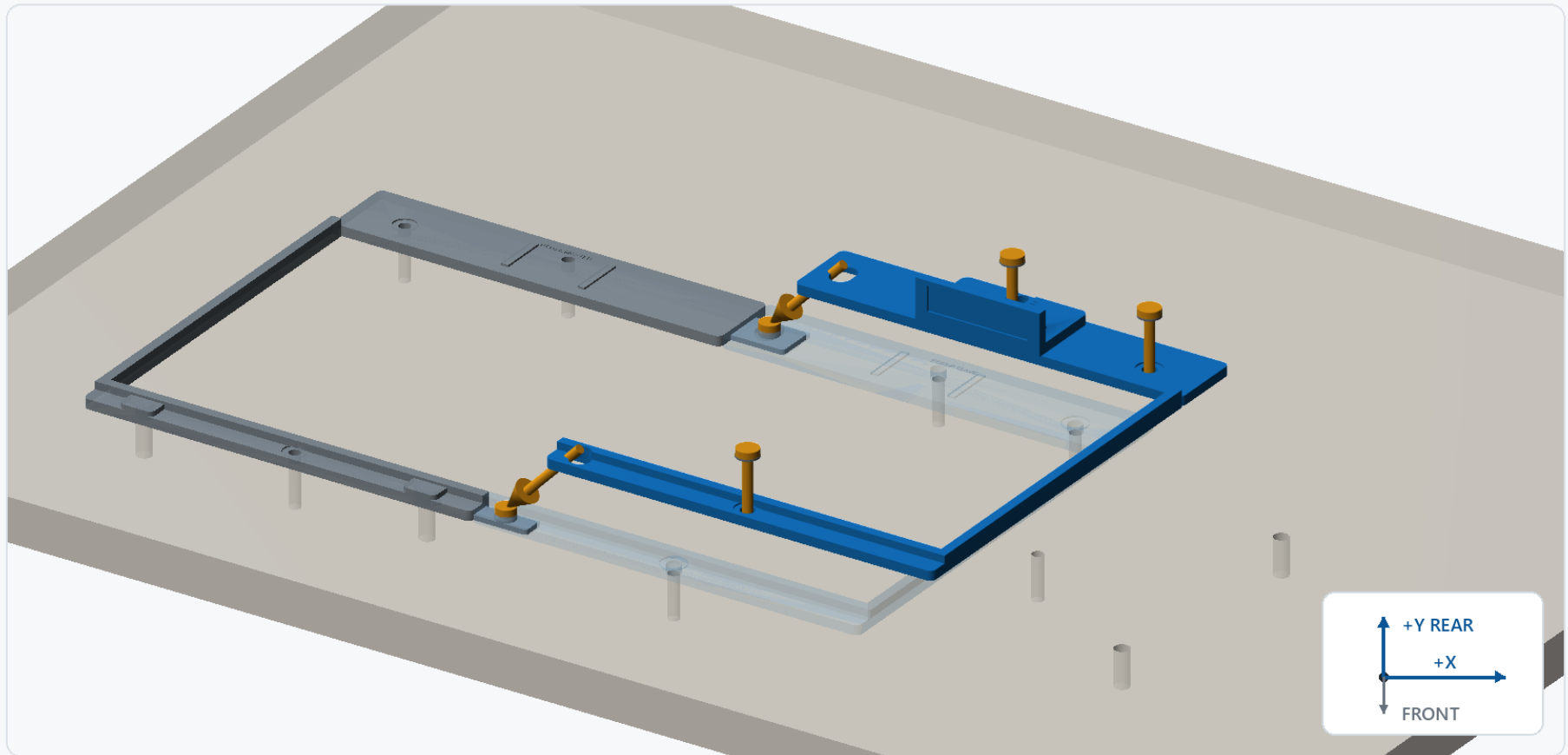
If a gate fails: stop, correct, and repeat this step

04 / 15 JOIN KEYBOARD SLAVE

04

JOIN KEYBOARD SLAVE

Engage the right station with the master's round post and Y-relieving slot.



1

Align both seam sockets before lowering;
no right-side board pins

2

Start front + rear M4/washer retainers
loosely

3

Seat padded slider before shared third M4;
all 3x M4s only clamp

[] STEP ACCEPTED

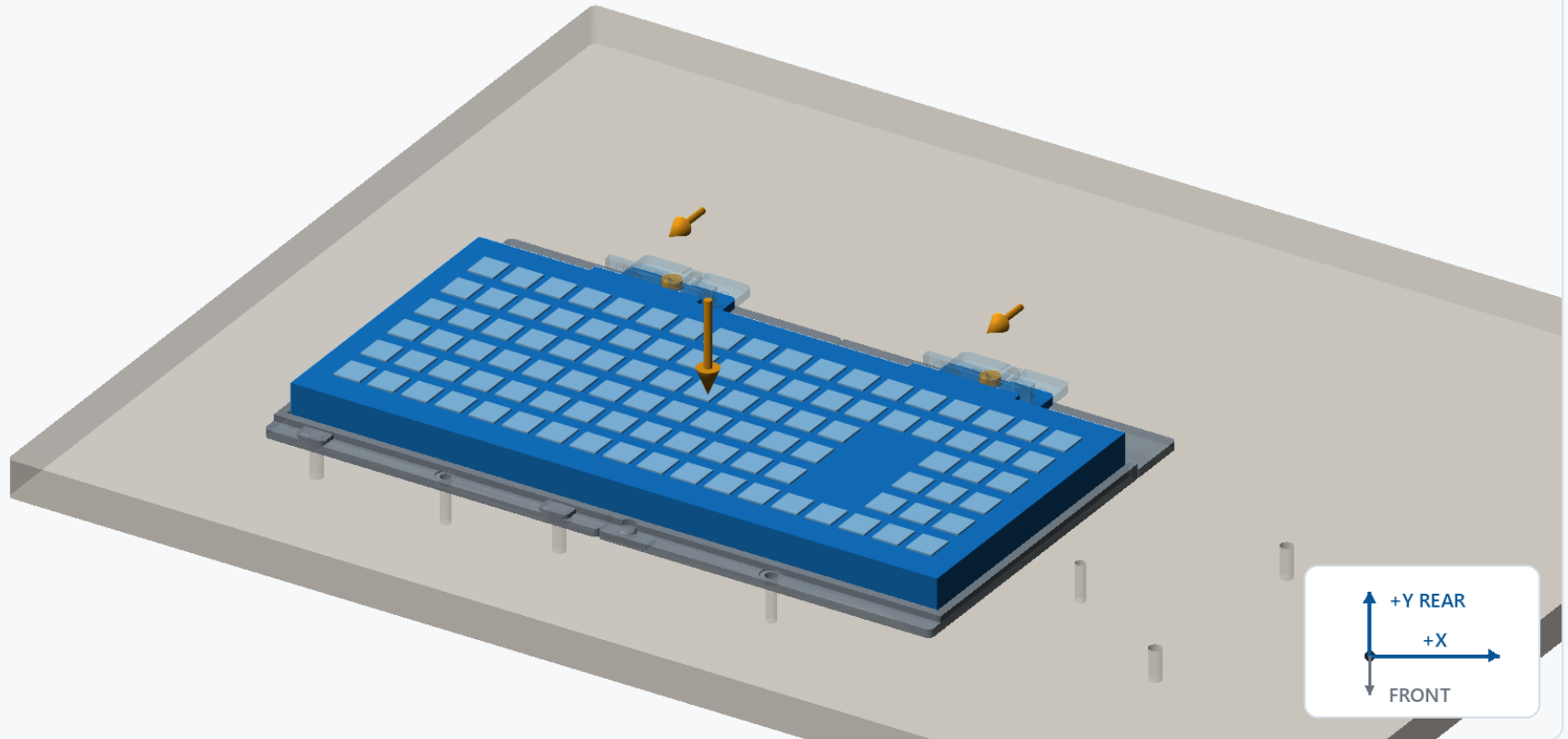
Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

05

LOAD AND CLAMP KEYBOARD

Place the keyboard on the board, then slide both rear clamps forward gently.



1

Keyboard rests directly on the sealed board

2

Rear face pads touch the case - no bending force

3

Third retainer passes through slider + station into board anchor

[] STEP ACCEPTED

Measurement / photo ID: _____

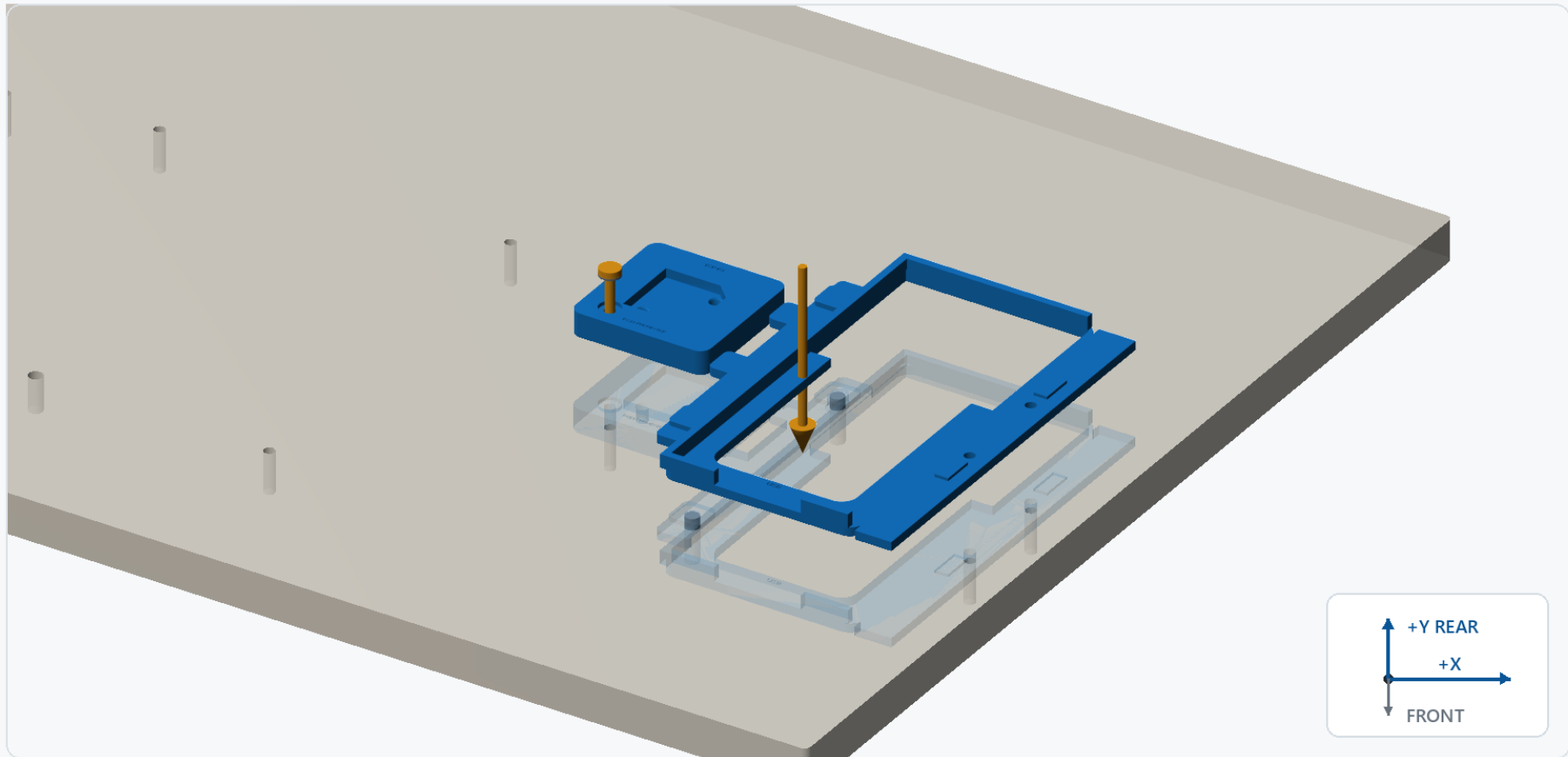
If a gate fails: stop, correct, and repeat this step

06 / 15 INSTALL PHONE AND TCP STATION

06

INSTALL PHONE + TCP STATION

Seat the combined station on its two locator pins before adding hardware.



1

2x locator pins: round socket fixes XY;
Y-slot relieves the axis

2

Install 1x TCP-side M4 + washer only

3

PT-HOLD-R1/R2 remain empty until the rail
is seated in Step 07

☐ STEP ACCEPTED

Measurement / photo ID: _____

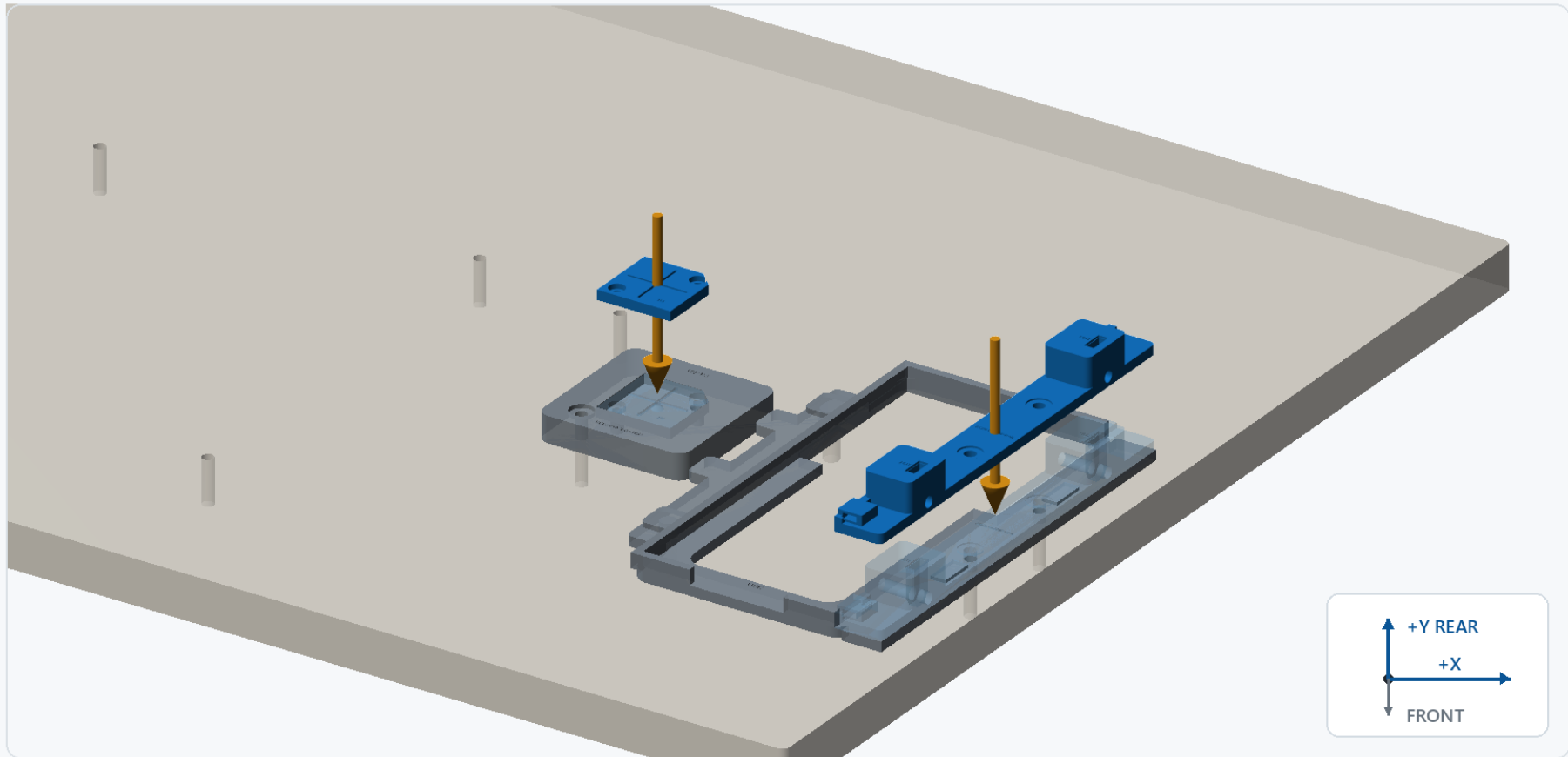
If a gate fails: stop, correct, and repeat this step

07 / 15 SEAT RAIL AND TCP CARTRIDGE

07

SEAT RAIL + TCP CARTRIDGE

Engage the rail keys first; place the keyed cartridge fully into its receiver.



1

Rail must sit flat on both locating keys

2

Clipped cartridge corner permits only one orientation

3

Do not insert the 2x shared rail screws until both parts are seated

[] STEP ACCEPTED

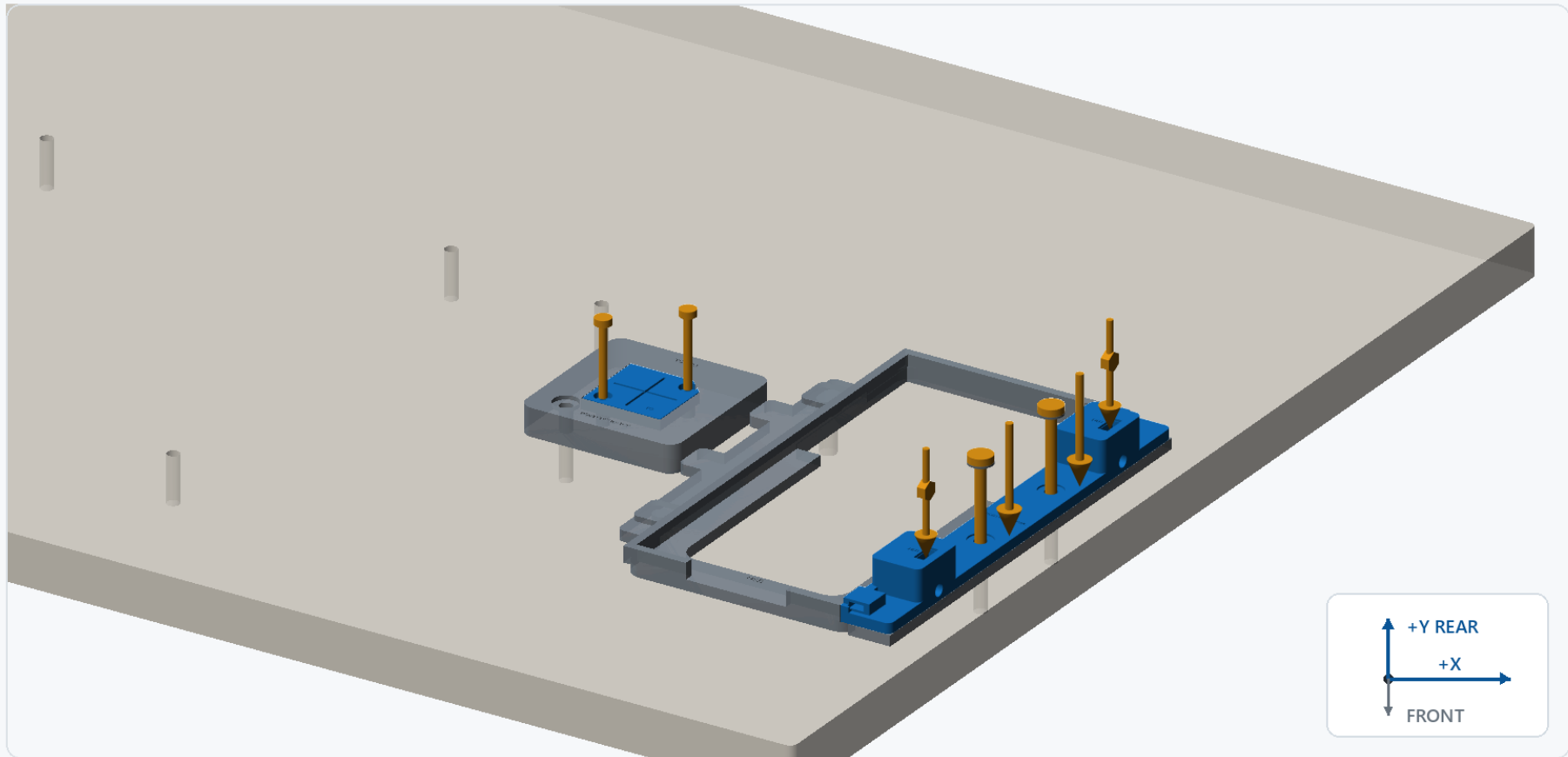
Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

08

INSTALL SERVICE HARDWARE

With the rail seated, add its shared board screws, clamp nuts and TCP screws.



1

Install 2x shared rail/station M4 screws + washers

2

Drop 2x M4 clamp nuts from the top

3

Install 2x M3 TCP screws; verify cartridge remains flush

☐ STEP ACCEPTED

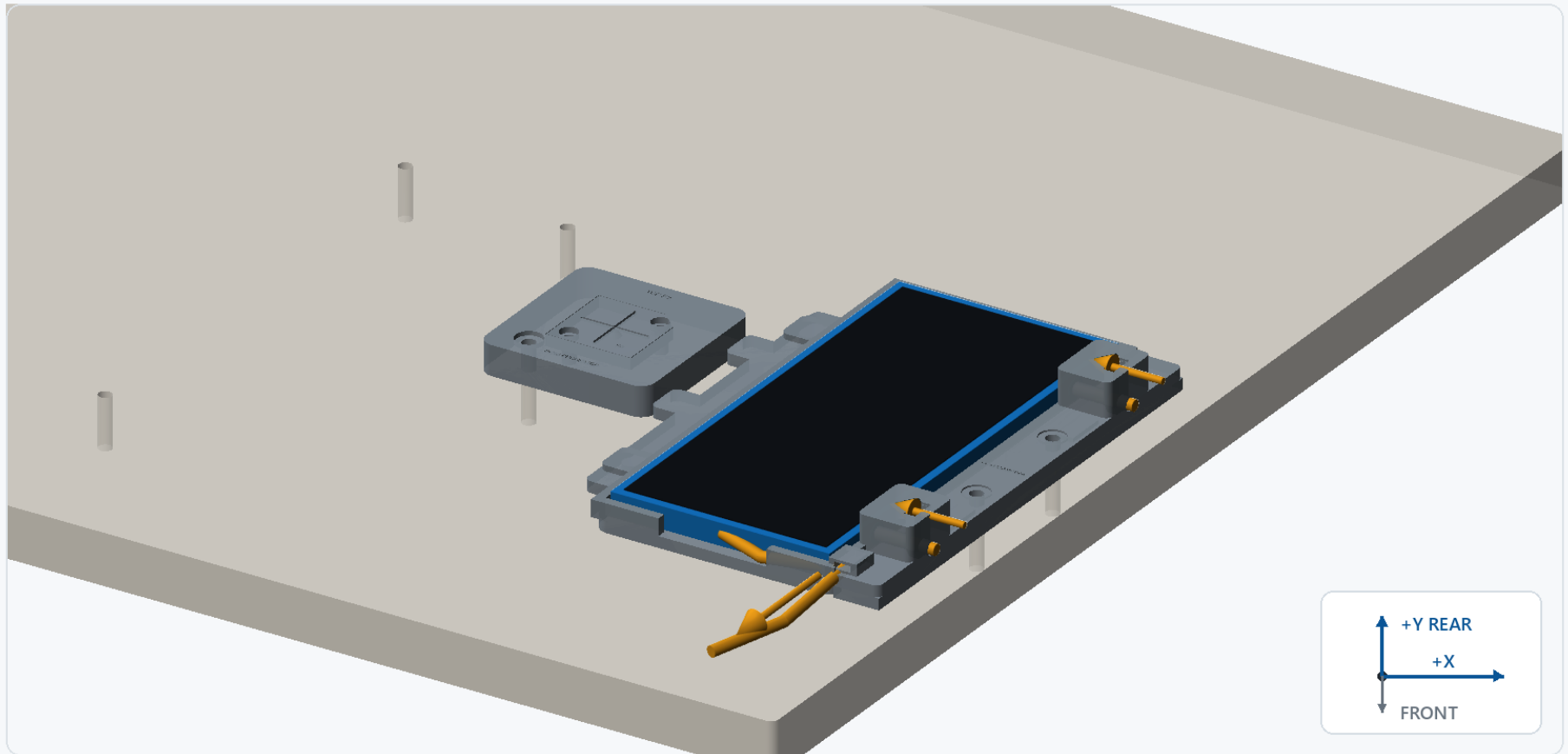
Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

09

LOAD PHONE + ROUTE CABLE

Seat the measured phone/case, retain the USB lead, then advance soft clamp tips.



1

USB plug and bend radius remain unstressed

2

Cable exits toward the front through the saddle

3

TPU tips avoid buttons, camera bump and screen edge

☐ STEP ACCEPTED

Measurement / photo ID: _____

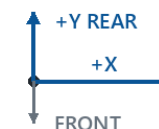
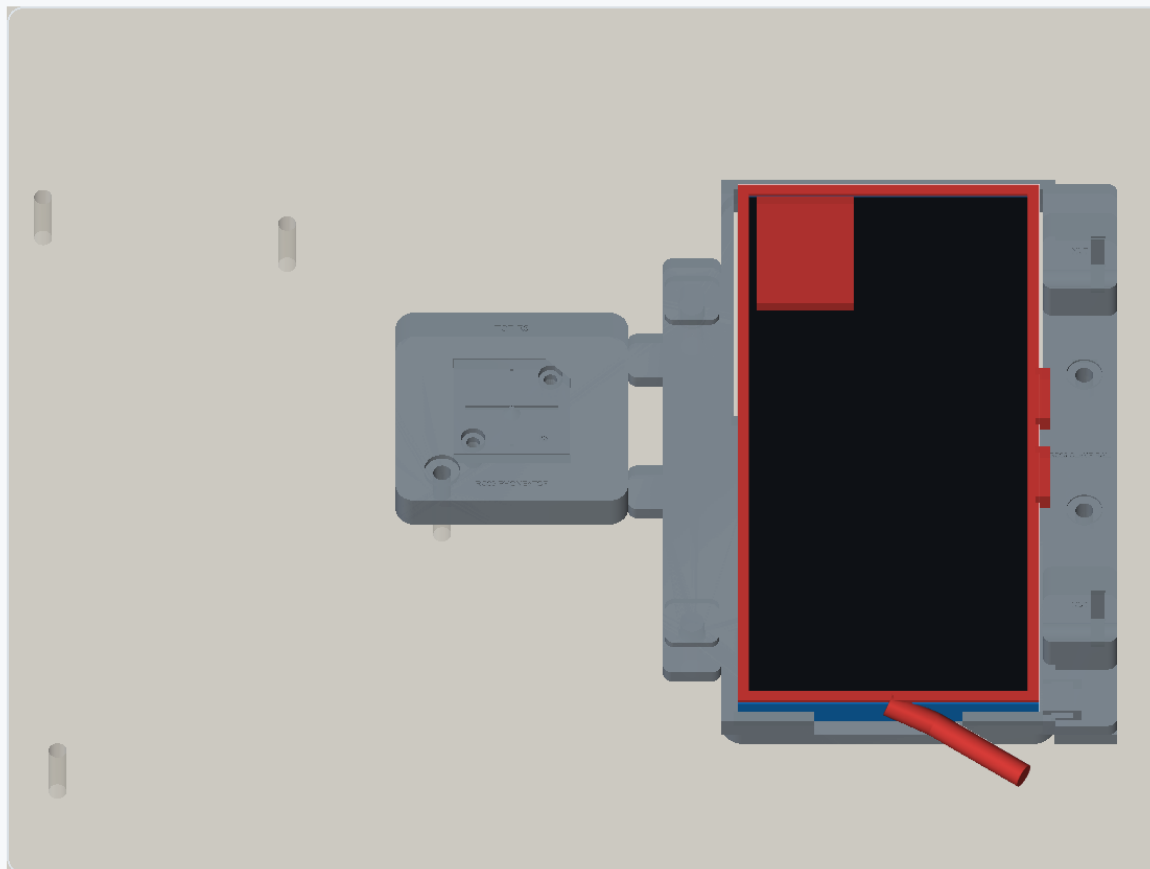
If a gate fails: stop, correct, and repeat this step

10

VERIFY PHONE NO-GO ZONES

RED = MEASURED NO-GO

Red zones are measurement reminders - not released positions - before either clamp is tightened.



1 Screen edge: soft tips remain below glass/bezel

2 Camera bump: measure actual case protrusion and keep clear

3 Side buttons: map power/volume positions and keep clear

4 USB: preserve connector body and measured bend radius

☐ STEP ACCEPTED

Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

11 / 15 TRANSFER TAG CENTERS AND DIRECTION

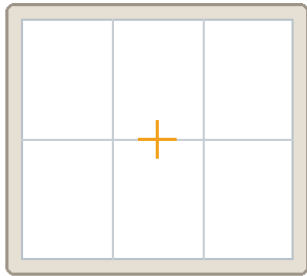
11

TRANSFER TAG CENTERS + DIRECTION

TEMPLATE → MARKS → SQUARE FRAME

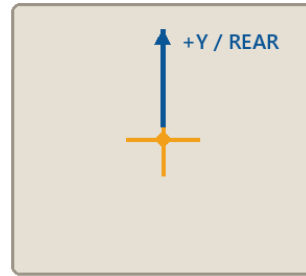
Register the 1:1 template, transfer center and +Y marks, then remove all paper.

A REGISTER TEMPLATE



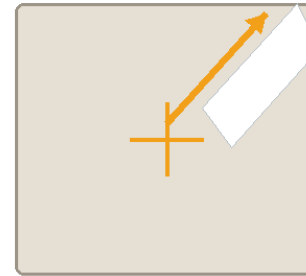
Tape at 100% / Actual Size

B TRANSFER MARKS



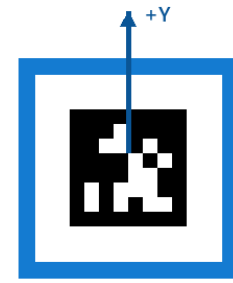
Transfer center AND direction

C REMOVE PAPER



Marks remain on sealed board

D FRAME + TAG



Square frame is removed after use

CONTROLLED TAG MAP

T0 / ID 0 center (47, 40) mm

T1 / ID 1 center (440, 40) mm

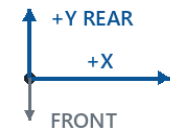
T2 / ID 2 center (47, 410) mm

T3 / ID 3 center (563, 410) mm

K0 / ID 4 center (324, 309) mm

P0 / ID 5 center (459, 309) mm

ALL TOP MARKS FACE +Y / REAR



1

Do not glue through the paper template

2

Logical labels map exactly to IDs 0-5 as shown

3

Use the transferred +Y mark to prevent a 90° tag rotation

[] STEP ACCEPTED

Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

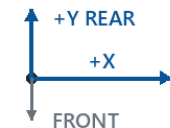
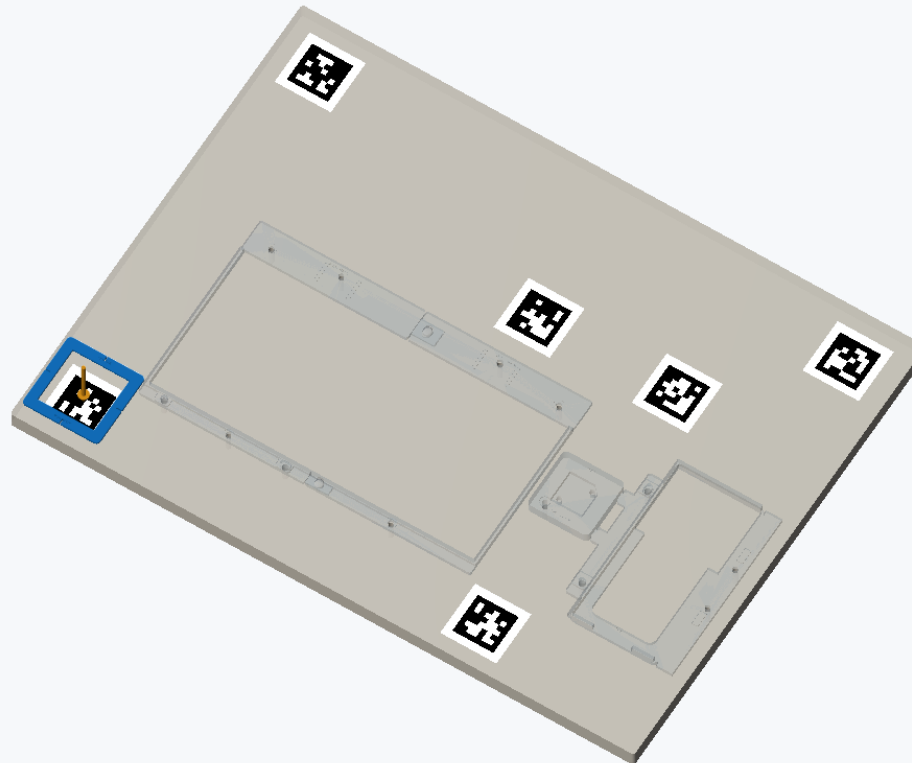
12 / 15 APPLY THE SIX APRILTAGS

12

APPLY THE SIX APRILTAGS

Place the square 55 mm frame on transferred marks; keep every tag top facing +Y.

SQUARE FRAME = TOOL - REMOVE IT



1

Apply directly to the cured sealed board

2

Repeat T0, T1, T2, T3, K0 and P0

3

Remove tool; measure center, yaw and optical plane Z

☐ STEP ACCEPTED

Measurement / photo ID: _____

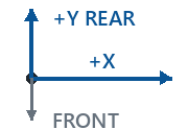
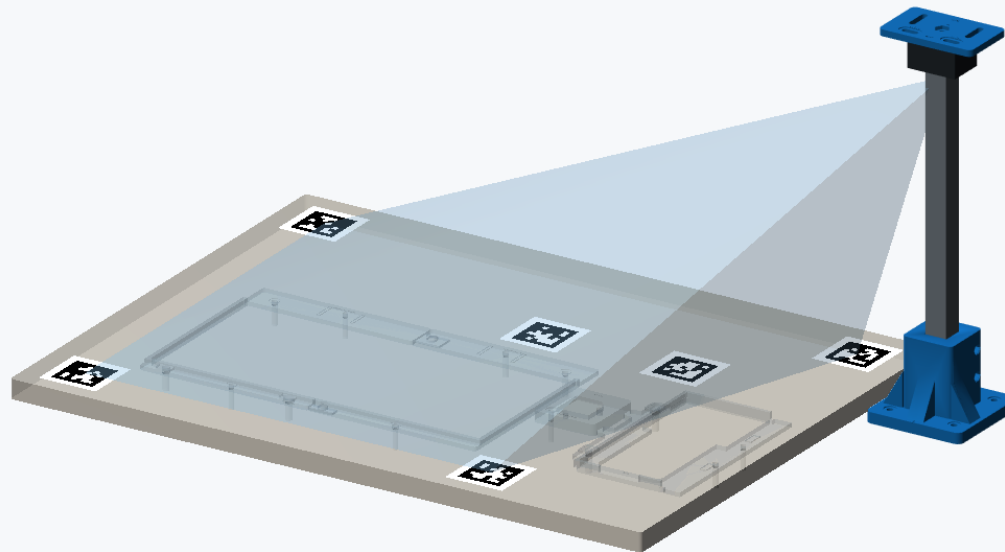
If a gate fails: stop, correct, and repeat this step

13

CAMERA ARCHITECTURE HOLD

FALLBACK SHOWN - NOT RELEASED

Intended arm-mounted camera is not yet defined. The image shows the fixed-mast fallback only - do not build Step 13.



1

STOP: exact camera, carrying link/frame and mount are required

2

Add camera mass/COM, moving-cable sweep, reach and collision proof

3

Define eye-in-hand calibration, timing and tag visibility by pose

☐ STEP ACCEPTED

Measurement / photo ID: _____

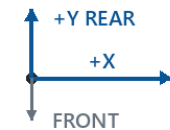
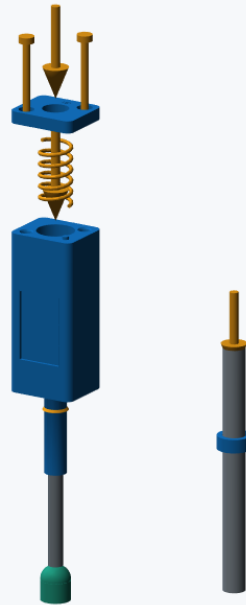
If a gate fails: stop, correct, and repeat this step

14

ASSEMBLE COMPLIANT TOOL

CONTACT TOOL - NOT A HARD STOP

Build and qualify each selected contact route one at a time, then calibrate its TCP.



1 Blue: configured body, keyed cap and selected adapter/collar

2 Orange: spring + 2x M3 cap screws

3 Select one or both routes; never install both route stacks simultaneously

4 After every tool change: verify travel, retention and recalibrate TCP

[] STEP ACCEPTED

Measurement / photo ID: _____

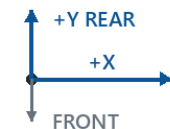
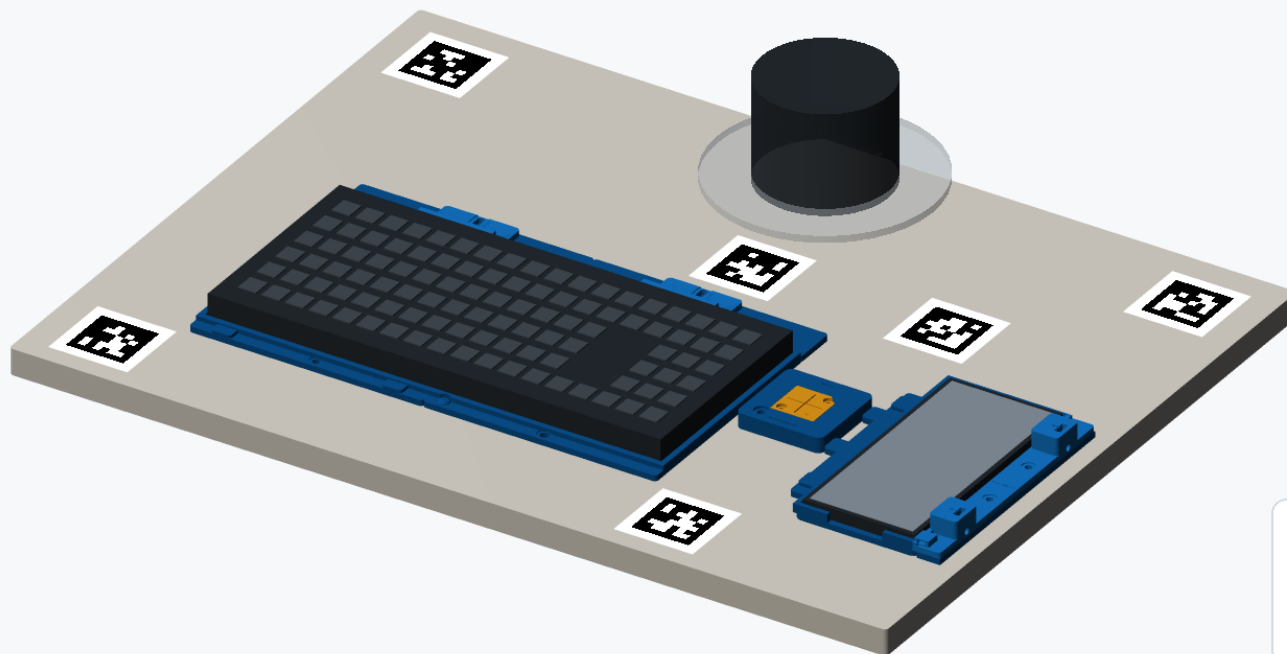
If a gate fails: stop, correct, and repeat this step

15

FINAL ASSEMBLY CHECK

RC03-INT-R1

Verify every station is seated, every device is gentle and every datum is visible.



1

Black rear cylinder = RoArm keep-out proxy, not released clamp CAD

2

Run 10x station remove/reinstall cycles; calibrate board + TCP

3

Run empty-cell motion before keyboard or phone contact

[] STEP ACCEPTED

Measurement / photo ID: _____

If a gate fails: stop, correct, and repeat this step

Dimensional and seating gates

Record actual values. A checkmark without a measured result is not acceptable evidence.

Gate	Check	Acceptance limit	Measured result
A01	Board local flatness	≤ 0.50 mm over any 300 mm span	Worst: _____ mm PASS [] FAIL []
A02	Board overall bow	≤ 1.50 mm over full board	Measured: _____ mm PASS [] FAIL []
A03	Station free-state corner lift	≤ 0.75 mm before clamping	Worst: _____ mm PASS [] FAIL []
A04	Keyboard support plane	≤ 0.40 mm variation	Range: _____ mm PASS [] FAIL []
A05	Phone support plane	≤ 0.30 mm variation	Range: _____ mm PASS [] FAIL []
A06	TCP cartridge seat	≤ 0.15 mm seat error	Measured: _____ mm PASS [] FAIL []
A07	Keyboard seam gap	≤ 0.40 mm	Measured: _____ mm PASS [] FAIL []
A08	Keyboard seam flush	≤ 0.25 mm	Measured: _____ mm PASS [] FAIL []
A09	Locator bores	15 mm blind depth; no breakout	Depths: _____ PASS [] FAIL []
A10	Locator projection	5.0 mm above board top	Pins: _____ mm PASS [] FAIL []
A11	Station retainers	0.25-0.35 N m; ≥ 5 mm engagement	Torque: _____ N m PASS [] FAIL []
A12	Screw termination	No bottoming or underside projection	[] All 9 pass PASS [] FAIL []

Evidence folder / report: _____ Witness: _____ Date: _____

Repeatability, vision, structure, and motion gates

Stop at the first failure. Correct the cause, repeat the affected build step, then repeat this gate.

Gate	Check	Acceptance limit	Measured result
B01	Station reinstall repeatability	10 cycles: X/Y ≤ 0.25 mm; yaw ≤ 0.20 deg; Z ≤ 0.20 mm	Worst: _____ PASS [] FAIL []
B02	Keyboard device pose	≤ 0.50 mm from recorded reference	Error: _____ mm PASS [] FAIL []
B03	Phone device pose	≤ 0.35 mm from recorded reference	Error: _____ mm PASS [] FAIL []
B04	Cartridge free play	≤ 0.15 mm	Measured: _____ mm PASS [] FAIL []
B05	Cartridge height range	≤ 0.10 mm across removals	Range: _____ mm PASS [] FAIL []
B06	Printed AprilTag tile	55.0 ± 0.2 mm; marker edge 40.0 ± 0.2 mm	Worst: _____ PASS [] FAIL []
B07	Tag placement	Center ≤ 0.50 mm; yaw ≤ 0.30 deg; edge lift ≤ 0.20 mm	Worst: _____ PASS [] FAIL []
B08	Vision detection	20 / 20 detections per tag	T0-T3/K0/P0: _____ PASS [] FAIL []
B09	Vision reprojection	≤ 1.0 px	Worst: _____ px PASS [] FAIL []
B10	Station proof loads	20 N lateral; 10 N functional; permanent shift ≤ 0.10 mm	Shift: _____ mm PASS [] FAIL []
B11	Board anchor proof	50 N axial for 60 s; no damage or shift	Worst: _____ PASS [] FAIL []
B12	Compliant tool travel	3-6 mm smooth travel; returns freely	Travel: _____ mm PASS [] FAIL []
B13	Empty-cell motion	No collision; measured clearance recorded	Min: _____ mm PASS [] FAIL []

Evidence folder / report: _____ Witness: _____ Date: _____

Final controlled startup and signoff

FIRST POWERED RUN

- ☐ E-stop function test passes before robot enable
- ☐ All devices and removable tools are out of the cell
- ☐ Speed and force limits are set to commissioning values
- ☐ Empty-cell path runs without collision
- ☐ Minimum measured clearance is recorded
- ☐ Camera sees T0-T3, K0, and P0 across required motion
- ☐ One device at a time is loaded and verified
- ☐ Cable slack loop clears the screen, rail, and robot envelope
- ☐ Final process path passes at controlled speed

RELEASE RECORD

Builder: _____

Witness: _____

Build date: _____ Location: _____

QIDI Plus4 serial: _____

Filament / lot: _____

Caliper ID: _____ Torque tool ID: _____

Force gauge ID: _____

Evidence folder: _____

Deviation record: _____

Final measured clearance: _____ mm

PHYSICAL RELEASE DECISION

I confirm that every acceptance gate was physically measured, passed, and linked to retained evidence.

RELEASED [] NOT RELEASED [] CONDITIONAL - DEVIATION ATTACHED []

Responsible signature: _____ Date: _____ Revision: RC03-INT-R1

Digital validation is not physical release. An unsigned guide remains UNRELEASED.

Archive this signed record with photos, measurement sheets, force data, camera evidence, and deviations.